Kota Mahathi Priya

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.22
Narayana Group of Institutions	Class XII (Telangana State board)	2023	97.70%
St.Anthony's High School	Class X (Telangana state board)	2021	100.00%

SCHOLASTIC ACHIEVEMENTS

- Achieved CQMS cadet rank in NCC

POSITIONS OF RESPONSIBILITY

Member, Eco Club (September 2026 - Present)

- Contributed to organizing the **Babul Eco Festival** and coordinating event activities.
- Supporting the planning of upcoming events and reaching out to potential sponsors for funding.

Team Lead, ChemE Cube (November 2025 - March 2026)

- Led a team to develop a working **CO capture prototype**.
- Designed, tested, and submitted the prototype for the **AIChE ChemE Cube competition**.

SKILLS

- **Programming:** C, Python, SQL, MATLAB
- **Machine Learning:** Supervised & Unsupervised Learning, Deep Learning, Computer Vision
- **Python Libraries:** NumPy, Pandas, Matplotlib, Seaborn, Streamlit, Scipy, Scikit-learn, Tensorflow
- **Engineering & Simulation:** Aspen, Solid Edge, Materials Studio

RELEVANT COURSES

- CH5900 - Computational Fluid Dynamics
- CH5720 - AI/ML Tool Chains for Chemical Engineers
- CH5010 - Advanced Numerical Methods
- CH5150 - Optimization Techniques-I
- CH5190 - Molecular Modelling of Catalytic Reactions
- CH5430 - Principles of Game Theory
- CH5400 - Introduction to Colloids and Interface Science

- CH5483 - Computer vision in Chemical and life Science

PROJECTS

Network Intrusion Detection using RBM

- Developed an RBM-based Network Intrusion Detection System to learn latent features from network traffic data.
- Built an **RBM-SVM hybrid classification pipeline** for detecting normal and anomalous network activity.
- Optimized model hyperparameters using RandomizedSearchCV to improve classification performance.
- Achieved **88.2% test accuracy** across **7,558 network traffic samples**.

DFT Study on Dissociation Behaviour of 2-Pyrone (Materials Studio, DMol³)

- Performed **computational analysis** of 2-pyrone dissociation using geometry optimization, vibrational frequency analysis, and transition-state search.
- Analyzed reactant, product, and transition-state energetics to characterize the reaction mechanism and energy barrier.
- Estimated a **1.61 eV activation energy** and analyzed vibrational modes to validate the transition state and reaction feasibility.

Synthesis of Cu Nanoparticles in Different Reactors (*Under Dr. Vamsi Vikram Gande, IIT Hyderabad)*

- Synthesizing **Cu nanoparticles** using different reactor configurations.
- Comparing the **properties and characteristics** of nanoparticles obtained under different reactor conditions.

CO Capture Prototype (ChemE Cube Competition, AIChE)

- Developed a CO capture prototype using **MEA-coated silicon pellets**.
- Designed and tested the prototype to evaluate its **CO capture performance**.

Solid-Liquid Extraction Simulation & Optimization (MT-2 Term Project)

- Developed a **Python simulation** for multistage solid-liquid extraction using numerical solvers.
- Developed an **ANN model** to predict solute removal and analyzed process parameters for extraction optimization.

EXTRACURRICULAR

- Enjoy playing badminton and practicing yoga.

- Interested in pencil sketching, singing, dancing, and cooking.
- Reading fiction, autobiographies, and mythology.

- Visited **Granules India Limited** to gain exposure to pharmaceutical manufacturing operations and processes.